1. Introduction of Computer

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Content

1. What is the Computer?

A **computer** is an electronic device that receives data, processes it according to instructions, and produces meaningful output.

Key Characteristics of a Computer:

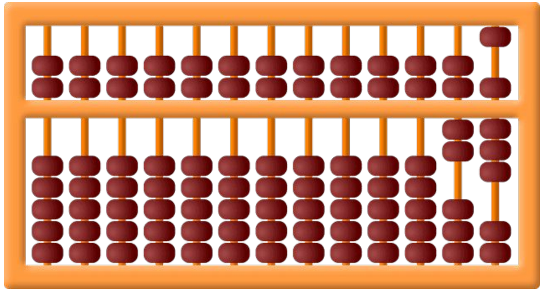
- **Speed:** Can perform millions of operations per second.
- **Accuracy:** Produces error-free results when instructions are correct.
- **Storage:** Can store large amounts of data.
- **Automation:** Works automatically once instructions are given.
- **Versatility:** Can perform many types of tasks.
- **Diligence:** Does not get tired or bored.

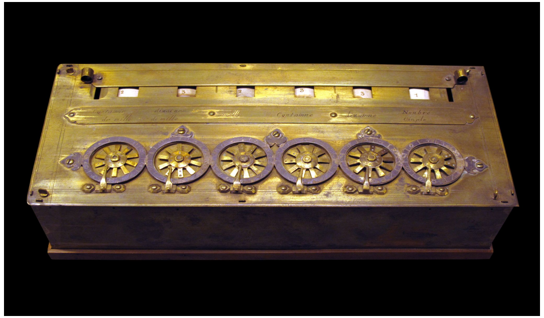
2. History of Computer

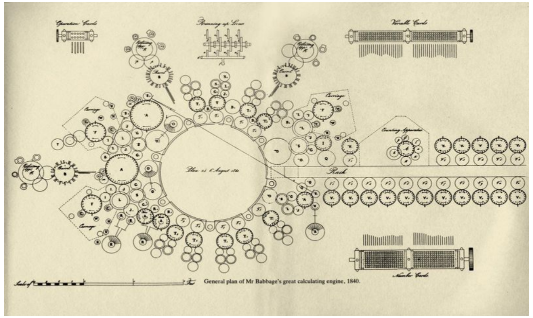
The history of computers is usually explained **generation-wise**, showing how computers evolved from simple calculating machines to powerful modern systems.

- Early Computing Devices (Before 1940)
- First Generation Computers (1940–1956) – Vacuum Tubes
- Second Generation Computers (1956–1963) – Transistors
- Third Generation Computers (1964–1971) – Integrated Circuits (ICs)
- Fourth Generation Computers (1971–Present) – Microprocessors
- Fifth Generation Computers (Present & Future) – AI & Quantum Computing

Early Computing Devices (Before 1940)







Key Devices

- **Abacus (c. 3000 BC)**

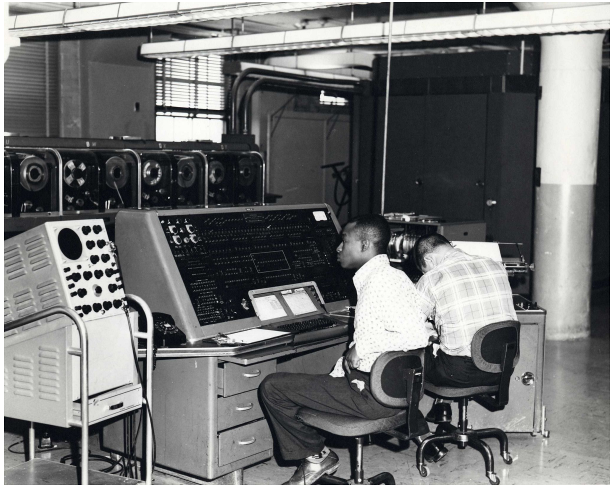
  First known calculating tool used for basic arithmetic.
- **Pascaline (1642)** – *Blaise Pascal*

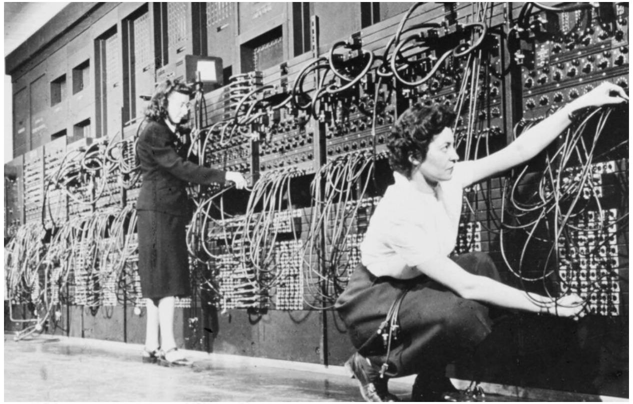
  Mechanical calculator for addition and subtraction.
- **Analytical Engine (1837)** – *Charles Babbage*

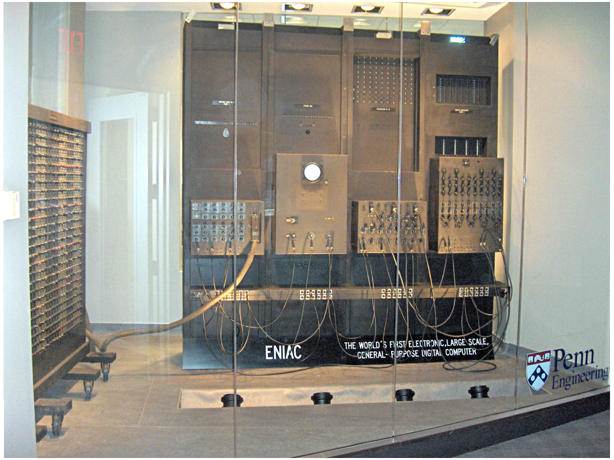
  Considered the **father of the computer**. It introduced ideas of:
  - Input
  - Processing
  - Memory
  - Output
- **Ada Lovelace**

  First computer programmer (wrote algorithms for the Analytical Engine).

First Generation Computers (1940–1956) – Vacuum Tubes







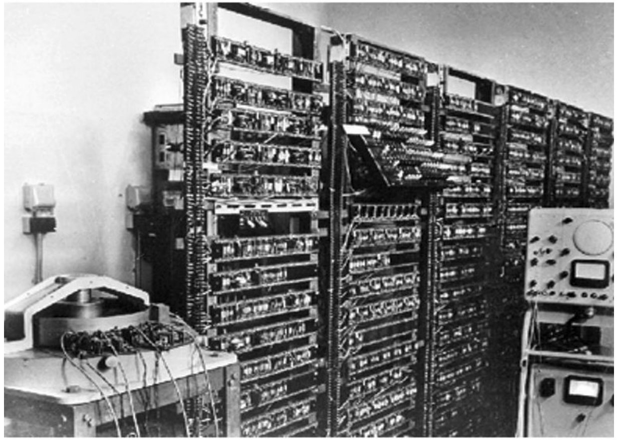
Characteristics

- Used **vacuum tubes**
- Very large in size (room-sized)
- Consumed huge electricity
- Generated a lot of heat
- Programming in **machine language**

Examples

- **ENIAC**
- **UNIVAC-I**

Second Generation Computers (1956–1963) – Transistors



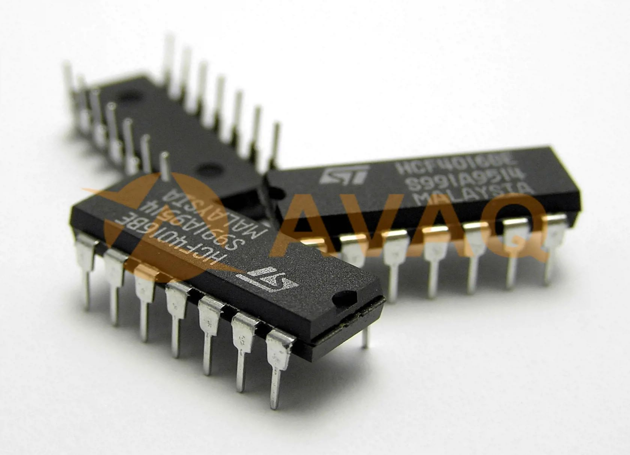
Characteristics

- Used **transistors** instead of vacuum tubes
- Smaller, faster, and more reliable
- Less power consumption
- Programming languages: **Assembly, FORTRAN, COBOL**

Example

- **IBM 1401**

Third Generation Computers (1964–1971) – Integrated Circuits (ICs)



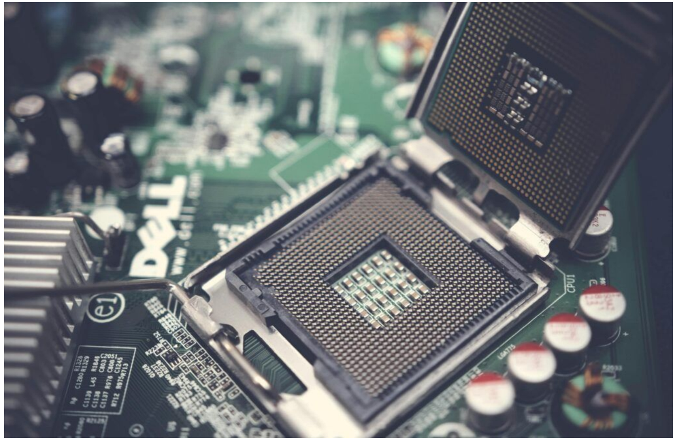
Characteristics

- Used **ICs (Integrated Circuits)**
- Much smaller and cheaper
- Faster processing
- Introduction of **operating systems**

Example

- **IBM System/360**

Fourth Generation Computers (1971–Present) – Microprocessors



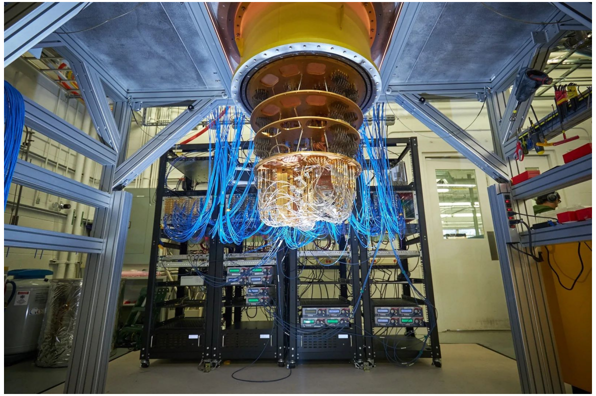
Characteristics

- Used **microprocessors**
- Birth of **personal computers** (PCs)
- High-speed processing
- Programming languages: **C, C++, Java**

Examples

- **Intel 4004**
- **IBM PC**
- **Apple Macintosh**

Fifth Generation Computers (Present & Future) – AI & Quantum Computing



Characteristics

- Based on **Artificial Intelligence**
- **Machine Learning & Neural Networks**
- **Quantum computing**
- Voice recognition & automation

Examples

- AI systems (ChatGPT, Siri)
- Quantum computers (IBM, Google)

2. History of Computer

Stage / Generation	Time Period	Main Technology Used	Key Features	Examples / Contributors
Early Computing Devices	Before 1940	Mechanical devices	Basic calculations, manual or mechanical operation	Abacus, Pascaline (Blaise Pascal), Analytical Engine (Charles Babbage), Ada Lovelace
First Generation	1940–1956	Vacuum Tubes	Very large size, high power consumption, heat generation, machine language programming	ENIAC, UNIVAC-I
Second Generation	1956–1963	Transistors	Smaller size, faster speed, less heat, assembly & high-level languages	IBM 1401
Third Generation	1964–1971	Integrated Circuits (ICs)	Increased reliability, multitasking, operating systems introduced	IBM System/360
Fourth Generation	1971–Present	Microprocessors	Personal computers, high speed, low cost, GUI-based systems	Intel 4004, IBM PC, Apple Macintosh
Fifth Generation	Present & Future	Artificial Intelligence, Quantum Computing	Machine learning, natural language processing, automation	AI systems, Quantum computers (IBM, Google)

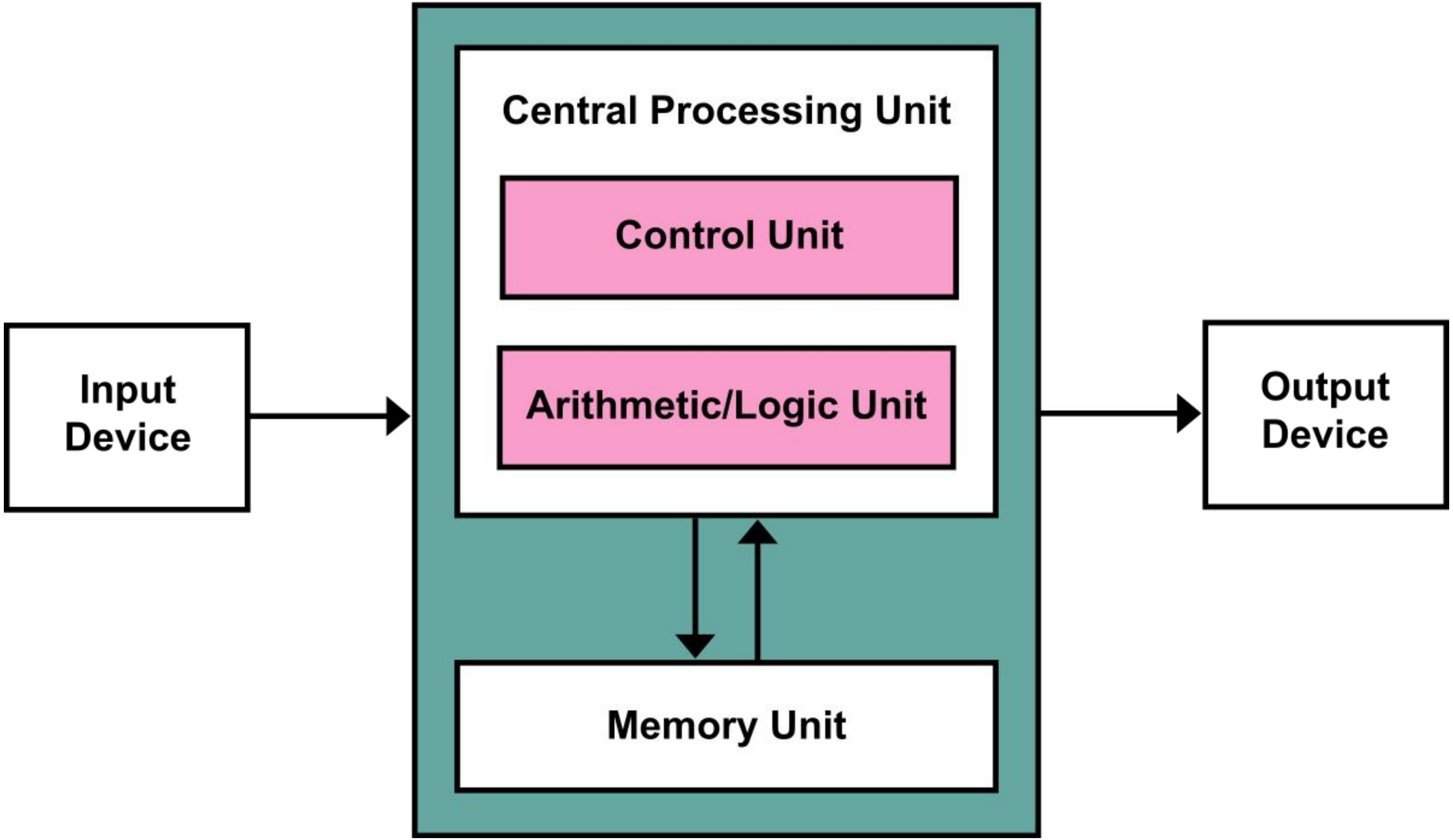
3. Basic Computer Organization

Every computer, from a smartwatch to a supercomputer, follows the same fundamental model known as the **Von Neumann Architecture**. It describes how the core parts of a computer work together.

It has four main units as follows:

1. Input Unit
2. Central Processing Unit
    a. Arithmetic Logic unit
    b. Control unit
3. Memory Unit
    a. Primary memory
    b. Secondary memory
4. Output Unit

3. Basic Computer Organization



3. Basic Computer Organization

1. Input Unit:

- **Function**: Accepts data and instructions from the outside world and sends them to the computer for processing.
- **Examples**: Keyboard, Mouse, Microphone, Scanner, Touchscreen.

3. Basic Computer Organization

2. Central Processing Unit (CPU):

- **Function**: This is the "brain" of the computer. It executes instructions, performs calculations, and manages data flow. It has two main parts:
  - **Control Unit (CU)**: It directs the operation of other components by managing the input/output devices and telling the memory and ALU how to respond.
  - **Arithmetic Logic Unit (ALU)**: It performs all mathematical calculations (+, -, *, /) and logical comparisons (>, <, =).

3. Basic Computer Organization

3. Memory (Storage):

This is where the computer stores data and instructions. It has two primary types:

- ***Primary Memory (RAM - Random Access Memory):***
    - **Function**: Temporarily holds the data and programs that are currently in use by the CPU. It is very fast, but volatile—meaning all data is erased when the computer is turned off.
    - More RAM = a bigger desk = more programs can run smoothly at the same time.

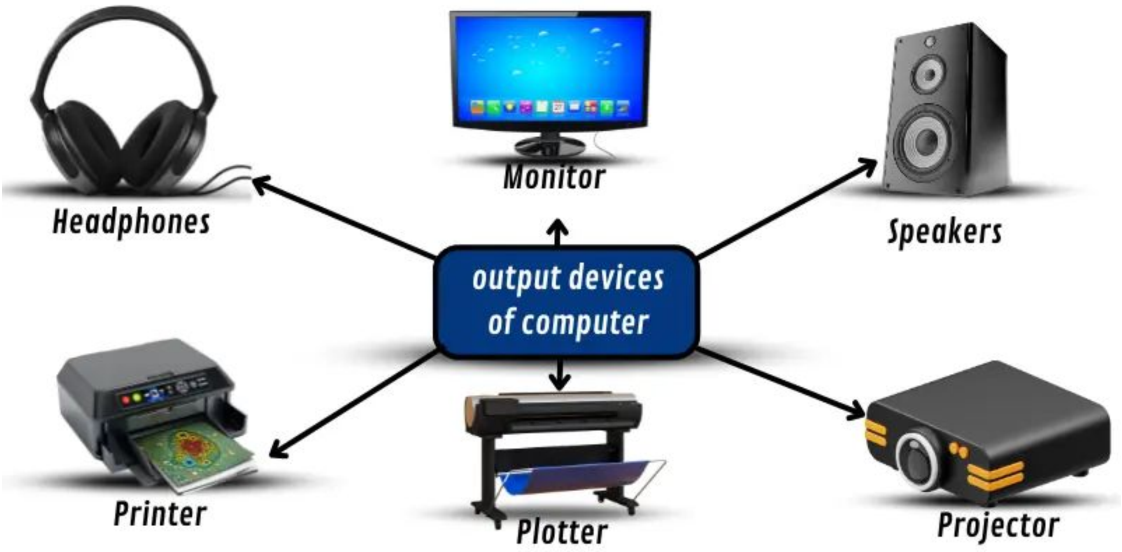
- ***Secondary Memory (Storage):***
    - **Function**: Permanently stores data (like OS, software, documents, photos) even when the computer is off. It is slower than RAM but non-volatile.
    - Examples: Hard Disk Drive (HDD), Solid State Drive (SSD), USB Flash Drive.

3. Basic Computer Organization

3. Basic Computer Organization

4. Output Unit:

- Function: Presents the processed data (results) to the user.
- Examples: Monitor, Printer, Speakers.



4. Data vs. Information

While often used interchangeably, Data and Information are fundamentally different concepts.

❖ **Data: The Raw Facts & Figures:**

- ***What it is***: Data is a collection of raw, unorganized facts, numbers, characters, or symbols. It has no meaning or context by itself.
- ***Characteristics***: It is unprocessed and discrete.
- ***Example***:
  - 32, 101, John, 95. Without context, these numbers and names are just data points.

❖ **Information: The Processed & Meaningful Data**

- ***What it is***: Information is data that has been processed, organized, structured, or presented in a given context to make it meaningful and useful.
- ***Characteristics***: It is processed, has context, and is useful for decision-making.
- ***Example*** (using the data above):
  - Context: Student Report Card
  - Information: "John scored 95 marks out of 100 in a class where the average temperature was 32°C and the room number was 101."
  - Now the data has context and tells a story. We have derived information about John's performance.

4. Data vs. Information

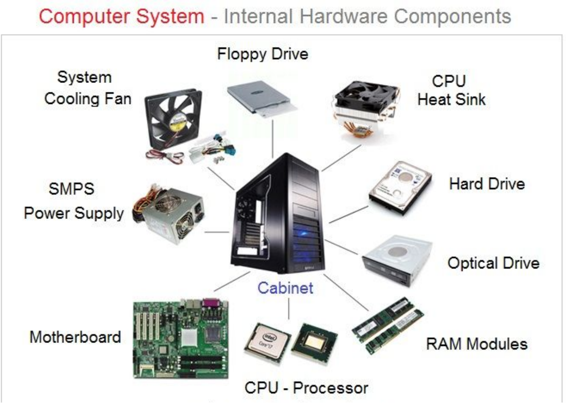
Aspect	Data	Information
Nature	Raw, unprocessed facts	Processed, organized data
Meaning	Has no meaning on its own	Has meaning and relevance
Dependency	Independent	Dependent on data
Usefulness	Not directly useful for decision-making	Directly used for making decisions
Example	`5, 10, 15`	`The average of the numbers is 10.`

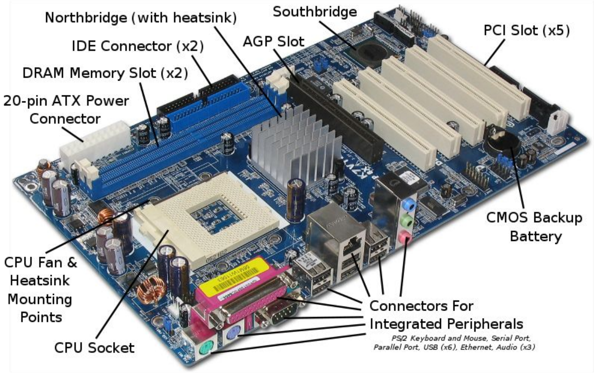
The Transformation:

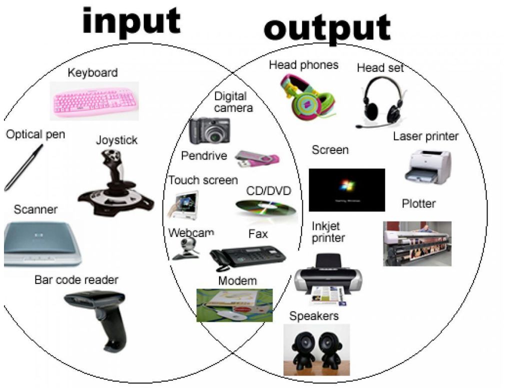
Data → [Processing by the Computer] → Information

This is the fundamental purpose of a computer: to take in raw Data, Process it according to instructions, and output useful Information.

5. Hardware







Hardware refers to the **physical parts** of a computer system that you can see and touch.

❑ **Examples of Hardware**

- **Input devices:** Keyboard, Mouse, Scanner
- **Processing unit:** CPU (Processor), Motherboard
- **Memory devices:** RAM, ROM
- **Storage devices:** Hard Disk, SSD, Pen Drive
- **Output devices:** Monitor, Printer, Speaker

❑ **Characteristics**

- Tangible (can be touched)
- Gets damaged physically
- Cannot work without software

6. Software

Software is a set of **programs and instructions** that tells the hardware what to do.

- ❑ **Types of Software**
  1. **System Software**
     - Controls hardware
     - Example: Windows, Linux, macOS
  2. **Application Software**
     - Helps users perform tasks
     - Example: MS Word, Excel, Browser
  3. **Programming Software**
     - Used to create programs
     - Example: GCC, VS Code, Python
- ❑ **Characteristics**
  - Intangible (cannot be touched)
  - Does not wear out physically
  - Needs hardware to run

7. Hardware and Software

Hardware vs Software (Comparison Table)

Feature	Hardware	Software
Nature	Physical	Logical
Can be touched	Yes	No
Examples	Keyboard, CPU	Windows, MS Word
Damage	Physical damage	Virus, bugs
Dependency	Needs software	Needs hardware

Relationship Between Hardware and Software

- Hardware is like the **body**
- Software is like the **brain**
- Both are **dependent on each other**

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Chapter 1